Change Point Detection in fMRI Data

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AN ESSAY

Submitted to the College of Liberal Arts and Sciences,

Wayne State University, Detroit, Michigan,

in partial fulfillment of the requirements for the degree of

MASTER OF ARTS IN APPLIED MATHEMATICS

April 2026

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Abstract

This report studies change point detection methods for high-dimensional functional magnetic resonance imaging (fMRI) data. The objective is to identify structural changes in multivariate brain activity signals and to compare how different statistical methods detect these changes. Because fMRI data consist of temporally ordered, high-dimensional observations, they provide a natural setting for evaluating both classical and nonparametric change point techniques.

The data are obtained from the OpenNeuro repository and consist of four-dimensional BOLD (blood-oxygen-level-dependent) imaging scans. After preprocessing and masking, the data are represented as a time-by-voxel matrix and subsequently reduced in dimension using principal component analysis (PCA). This produces a lower-dimensional multivariate time series that captures the dominant patterns of variation in brain activity.

Several change point detection methods are applied, including classical segmentation procedures and nonparametric approaches. Particular emphasis is placed on graph-based methods, which construct a similarity graph on the observations and detect changes based on the structure of the resulting graph.

A modified graph-based approach is introduced in which a temporal penalty is incorporated into the distance function used to construct the graph. This modification encourages connections between observations that are both similar in feature space and close in time. The results show that, while the original method identifies a highly significant change point near the end of the scan, the temporally weighted method reveals an alternative and stable change point earlier in the time series for sufficiently large values of the tuning parameter.

These findings demonstrate that incorporating temporal structure into graph-based methods can lead to different and potentially more interpretable segmentations of high-dimensional time series data. The results highlight the importance of method selection and model assumptions when analyzing complex multivariate signals such as fMRI data.

Contents

1 Introduction

This report studies change point detection methods for high-dimensional functional MRI (fMRI) data. The main objective is to understand how different statistical methods identify structural changes in brain activity over time, and to compare the types of changes each method is most sensitive to. Because fMRI data are multivariate, noisy, and temporally ordered, they provide a useful setting for comparing classical and nonparametric change point methods.

2 Motivation

2.1 Background on fMRI

Functional MRI measures changes in blood oxygenation over time, commonly referred to as the BOLD signal. Each voxel represents a small region of the brain, and the signal recorded at each voxel changes over time as neural activity and blood flow vary. Because an fMRI scan records many voxels simultaneously across many time points, the resulting data are high-dimensional and naturally ordered in time.

This makes fMRI data a useful setting for change point detection. If the underlying brain activity pattern changes because of a task, stimulus, or shift in cognitive state, then the distribution of the observed multivariate time series may also change. Different change point methods may detect different aspects of these changes, such as shifts in mean level, variance, dependence structure, or more general distributional differences.

2.2 Data Source

The data used in this project were obtained from the OpenNeuro repository, an open platform that hosts publicly available neuroimaging datasets. OpenNeuro provides datasets organized according to the Brain Imaging Data Structure (BIDS) standard, which defines a consistent folder structure and metadata format for neuroimaging experiments.

The dataset used in this study corresponds to the OpenNeuro dataset

Targeting Dynamic Facial Processing Mechanisms in Superior Temporal Sulcus Using fMRI Neurofeedback.

The dataset is available at

`https://openneuro.org/datasets/ds004141`

and is identified by the DOI

`doi:10.18112/openneuro.ds004141.v1.0.5`

This dataset contains functional MRI recordings collected during a neurofeedback experiment designed to study facial processing mechanisms in the superior temporal sulcus. The data were collected from multiple participants and include both anatomical MRI scans and functional BOLD imaging runs.

The dataset was located by searching the OpenNeuro repository for publicly available functional MRI experiments involving neurofeedback and task-based brain activity.

2.3 Dataset Structure

The dataset follows the Brain Imaging Data Structure (BIDS) format, which organizes neuroimaging data into standardized folders for subjects, sessions, anatomical scans, and functional scans.

The dataset contains 20 participants divided into experimental and sham groups. Participant-level metadata include age, sex, and group assignment.

For each subject, the dataset includes anatomical MRI scans and multiple functional MRI runs. The primary files used in this project are the functional BOLD scans stored as compressed NIfTI files with filenames ending in `_bold.nii.gz`. Event timing files are also provided for each functional run.

2.4 Functional MRI Measurements

The primary data used in this project are functional magnetic resonance imaging (fMRI) scans that measure brain activity over time using the blood-oxygen-level-dependent (BOLD) signal. The BOLD signal reflects changes in blood oxygenation that occur when neurons in a region of the brain become active. As neural activity increases, local blood flow and oxygenation also change, which produces measurable differences in MRI signal intensity.

Each functional scan is stored as a four-dimensional image file in NIfTI format with the structure

$$x \times y \times z \times t,$$

where the first three dimensions represent spatial location within the brain and the final dimension represents time. Each spatial unit in the image is called a voxel, which represents a small volume of brain tissue. During an fMRI experiment, the scanner repeatedly records brain volumes over time, producing a time series of measurements at each voxel.

For the functional run analyzed in this study, the dimensions of the data were

```
64 x 64 x 33 x 300
```

This indicates that each recorded brain volume consists of a grid of $64 \times 64 \times 33$ voxels, and that a total of 300 volumes were collected during the scan. Consequently, each voxel contains a time series of 300 measurements describing how the BOLD signal at that spatial location evolves during the experiment.

Because an fMRI scan records signals from thousands of voxels simultaneously, the resulting dataset is extremely high-dimensional. In the present dataset, the functional run used in the analysis contained over one hundred thousand voxel time series. These time series form a multivariate signal that evolves over time and can be studied using statistical methods designed to detect structural changes in high-dimensional data.

2.5 Construction of the Time Series Used in This Study

The raw functional MRI data were stored as four-dimensional NIfTI image files. To prepare the data for statistical analysis, the imaging data were first converted into a matrix representation in which rows correspond to time points and columns correspond to voxel signals.

The NIfTI file for a selected subject and experimental run was read into `R` as a four-dimensional array with dimensions

$$64 \times 64 \times 33 \times 300.$$

The first three dimensions correspond to spatial location in the brain, while the fourth dimension corresponds to time. Each element of this array represents the measured BOLD signal intensity at a particular voxel and time point.

Because large portions of the image correspond to background outside the brain, a simple masking procedure was used to remove voxels with zero signal. The mask was constructed using the first recorded brain volume, and only voxels with nonzero intensity were retained for further analysis.

After masking, the data were reshaped into a matrix in which each column represents the time series for a single voxel and each row represents a time point in the scan. For the selected run, this resulted in a matrix with

$$300 \text{ time points} \times 133{,}056 \text{ voxel signals.}$$

This matrix representation forms the basis for subsequent statistical analysis.

Because the number of voxel signals is extremely large, direct multivariate analysis is computationally challenging. To reduce the dimensionality of the data while preserving the dominant temporal structure, principal component analysis (PCA) was applied to the standardized voxel time series. PCA provides a lower-dimensional representation of the multivariate signal by projecting the data onto directions of maximal variance.

In this study, the first several principal components were used to construct a reduced representation of the brain activity time series. This lower-dimensional representation was then used as the input for the change point detection methods described in the following section.

3 Methods

3.1 Change Point Detection Framework

The objective of this study is to identify structural changes in a multivariate time series derived from functional MRI data. After preprocessing and dimensionality reduction, the data consist of a sequence of high-dimensional observations describing the evolution of brain activity over time.

A change point is defined as a time index at which the statistical properties of the observed data change. Depending on the method used, this change may correspond to a shift in mean level, variance, dependence structure, or the full probability distribution of the observations.

Detecting such changes can reveal time points where the underlying brain activity pattern shifts during the experiment.

3.2 Mathematical Formulation of the Change Point Problem

Let

$$X \in \mathbb{R}^{T \times p}$$

denote the multivariate time series obtained after preprocessing the fMRI data, where T represents the number of time points and p represents the number of voxel signals. In the present dataset, $T = 300$ and $p = 133{,}056$.

Each row $X_t \in \mathbb{R}^p$ represents the vector of voxel measurements observed at time point t.

A change point is defined as a time index τ at which the probability distribution of the observations changes. More formally, the sequence of observations can be written as

$$X_1, X_2, \ldots, X_T$$

where

$$X_1, \ldots, X_\tau \sim F_1$$

and

$$X_{\tau+1}, \ldots, X_T \sim F_2,$$

with $F_1 \neq F_2$.

The goal of change point detection is to estimate the unknown change point location τ by identifying where the statistical properties of the observations shift. Depending on the method used, this change may correspond to differences in the mean, variance, dependence structure, or the full multivariate distribution of the observations.

Because the dimensionality p of the voxel signals is extremely large, the analysis is performed on a lower-dimensional representation obtained through principal component analysis. Let

$$Z \in \mathbb{R}^{T \times k}$$

denote the reduced representation of the data obtained from the first k principal components. The change point detection methods described in the following sections are applied to this reduced multivariate time series.

3.3 Methods Considered

Several change point detection methods were applied to the reduced multivariate time series obtained after preprocessing and dimensionality reduction. These methods differ in the type of statistical changes they are designed to detect and in the assumptions they make about the underlying signal.

3.3.1 PELT

The Pruned Exact Linear Time (PELT) algorithm is a classical change point detection method designed to identify multiple change points in a time series by minimizing a penalized cost function. The method seeks a segmentation that minimizes

$$\sum_{i=1}^{m+1} C(y_{(\tau_{i-1}+1):\tau_i}) + \beta m,$$

where $C(\cdot)$ is a cost function measuring within-segment variability, m is the number of detected change points, and β is a penalty term that controls model complexity. In this study, PELT was applied to a one-dimensional summary of the data obtained from the first principal component.

3.3.2 Binary Segmentation

Binary segmentation is a recursive procedure that detects change points by repeatedly splitting the time series into two segments. At each step, the method identifies the location that maximizes a test statistic measuring the difference between the two segments. The procedure is then applied recursively to the resulting subsegments. Although computationally efficient, binary segmentation may miss closely spaced change points because it relies on a sequence of binary splits.

3.3.3 Energy Method

The rank-energy approach is a nonparametric method designed to detect general distributional changes in multivariate data. The method first applies a rank transformation to each coordinate of the observations to remove scale effects and produce a distribution-free representation. A scan statistic based on the energy distance between two segments is then computed across candidate split locations. The estimated change point corresponds to the location that maximizes this statistic.

3.3.4 Kernel / Maximum Mean Discrepancy Method

Kernel-based change point detection methods measure differences between probability distributions using reproducing kernel Hilbert space representations. In particular, the Maximum Mean Discrepancy (MMD) statistic compares two sets of observations by measuring the distance between their kernel mean embeddings. For candidate split points τ, the statistic measures the discrepancy between the distributions of observations before and after τ. Large values of the statistic indicate a potential change in the distribution of the time series.

3.3.5 Graph-Based Edge Count Method

Graph-based change point detection methods construct a similarity graph on the observations and examine how edges connect points across different time segments. In this study, a minimum spanning tree was constructed using pairwise distances between time points in the reduced representation Z. The edge-count statistic measures the number of edges that connect observations from different segments. Significant deviations from the expected edge structure under homogeneity indicate the presence of a change point.

3.4 Temporally Weighted Graph Construction

The standard graph-based method constructs a similarity graph on the observations using a distance based solely on feature similarity. In particular, after dimensionality reduction via principal component analysis (PCA), each time point is represented by a vector $Z_t \in \mathbb{R}^d$, and the distance between two observations is defined as

$$d_{\text{orig}}(i, j) = 1 - \text{cor}(Z_i, Z_j),$$

where $\text{cor}(\cdot, \cdot)$ denotes the sample correlation between two vectors.

A limitation of this approach is that it ignores the temporal ordering of the observations. As a result, the constructed graph may connect time points that are similar in feature space but far apart in time, potentially obscuring meaningful temporal structure in the data.

To address this, we introduce a modified distance that incorporates a temporal penalty:

$$d_\lambda(i, j) = \big(1 - \text{cor}(Z_i, Z_j)\big) + \lambda \frac{|i - j|}{n},$$

where n is the number of time points and $\lambda \geq 0$ is a tuning parameter controlling the influence of temporal proximity.

The additional term penalizes connections between observations that are far apart in time, encouraging the resulting graph to favor edges between both similar and temporally adjacent observations. When $\lambda = 0$, the method reduces to the original graph-based approach. As λ increases, temporal structure plays a more dominant role in determining the graph.

The modified distance is used to construct a weighted complete graph, from which a minimum spanning tree (MST) is obtained. Change point detection is then performed using the same graph-based scan statistics applied to the MST. This allows for a direct comparison between the original and temporally weighted methods.

4 Results

The graph-based change point detection method was applied to the reduced multivariate time series obtained from the fMRI data. The analysis revealed a highly significant change point at time index

$$\tau = 267.$$

The corresponding scan statistic attained a large value, with an associated p-value effectively equal to zero, providing strong evidence against the null hypothesis of no change in distribution.

4.1 Scan Statistic

Figure 1 displays the scan statistic as a function of time. The statistic exhibits a clear maximum at $t = 267$, indicating a candidate change point. The sharp peak suggests a well-defined structural change in the underlying signal.

figures/scan_statistic.png

Figure 1: Graph-based scan statistic over time with detected change-point at $t = 267$.

4.2 PCA Trajectory

To visualize the evolution of the multivariate signal, the data were projected onto the first two principal components. Figure 2 shows the trajectory of the observations in this reduced space, colored according to time.

The trajectory reveals a gradual evolution of the signal followed by a transition into a distinct region of the feature space toward the end of the scan. The late time points form a tight cluster that is clearly separated from earlier observations, supporting the presence of a structural change.

figures/pca_trajectory.png

Figure 2: Trajectory of the time series in the first two principal components, colored by time. Later observations cluster in a distinct region.

4.3 Graph Structure

The minimum spanning tree constructed from the PCA representation exhibited extremely few edges connecting observations before and after the detected change point. In particular, only a single edge in the tree connected points across the boundary at $t = 267$.

This indicates that the two segments of the time series occupy nearly disjoint regions of the feature space. Such a near-disconnection provides strong geometric evidence of a distributional change in the multivariate signal.

4.4 Effect of Temporal Weighting

To evaluate the impact of incorporating temporal information into the graph construction, the temporally weighted distance

$$d_\lambda(i,j) = \big(1 - \mathrm{cor}(Z_i, Z_j)\big) + \lambda \frac{|i-j|}{n}$$

was computed for a range of values of the tuning parameter λ. For each value, a minimum spanning tree (MST) was constructed and the generalized edge-count scan statistic was used to estimate the change-point location.

The results are summarized in Table 1.

λ	Estimated Change-Point ($\hat{\tau}$)	p-value
0.00	267	9.29×10^{-62}
0.10	267	9.23×10^{-62}
0.25	267	9.35×10^{-62}
0.50	154	4.28×10^{-63}
1.00	153	4.38×10^{-63}

Table 1: Estimated change-point and p-values for varying values of the temporal weighting parameter λ.

For small values of λ, the detected change-point remains unchanged at $\hat{\tau} = 267$, indicating that weak temporal penalization does not affect the dominant structure of the graph. However, as λ increases beyond a critical threshold (approximately $\lambda = 0.5$), the estimated change-point shifts substantially and stabilizes near $\hat{\tau} \approx 154$.

This transition demonstrates that incorporating temporal structure alters the connectivity of the graph in a meaningful way. While the original method connects observations based solely on feature similarity, the temporally weighted method discourages long-range connections and favors edges between temporally adjacent observations. As a result, a different and stable segmentation of the time series emerges.

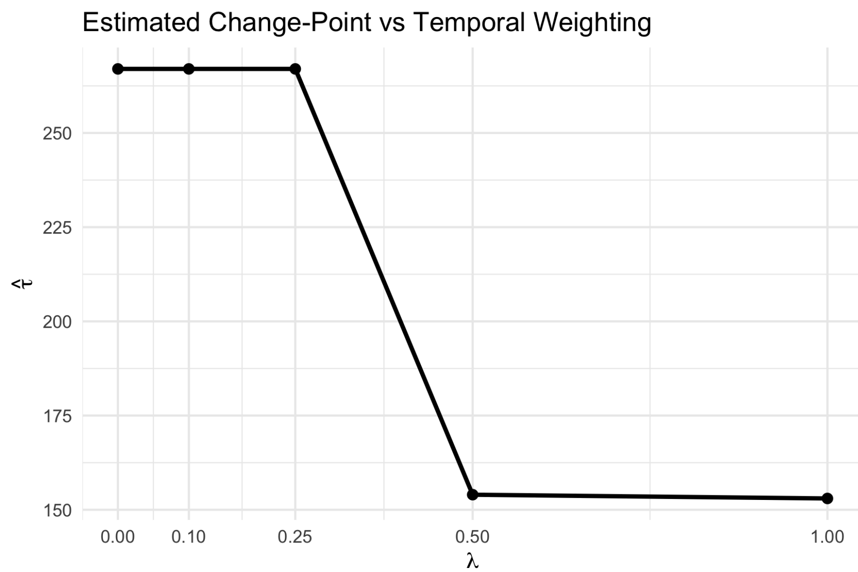

Figure 3: Estimated change-point location as a function of the temporal weighting parameter λ. A transition occurs near $\lambda = 0.5$, after which the detected change-point stabilizes.

Notably, the alternative change-point remains highly statistically significant, indicating that it reflects a strong structural feature of the data rather than noise. These results highlight that incorporating temporal information into graph-based change point detection can reveal alternative and potentially more interpretable change-points.

4.5 PCA Heatmap

Figure 4 displays a heatmap of the first several principal components over time. Each row corresponds to a principal component, and each column corresponds to a time point.

The heatmap reveals clear temporal structure in the data. Several principal components exhibit noticeable changes in magnitude and pattern near the detected change point. In particular, a shift in the behavior of multiple components is visible in the later portion of the time series.

This visualization provides additional support for the presence of a structural change and illustrates how the underlying multivariate signal evolves over time.

4.6 Multi-Subject Analysis

To examine whether the effect of temporal weighting persists across subjects, the graph-based change point analysis was applied to a collection of subjects using both the original method and the temporally weighted method with $\lambda = 0.5$. For each subject, the generalized edge-count statistic was used to estimate the change-point location.

The original method produced highly variable change-point estimates across subjects, while the temporally weighted method yielded estimates that were more concentrated, with many subjects having detected change-points near a common location. This pattern is shown in Figure 5, which compares the original and temporally weighted change-point estimates for all subjects. The dashed line represents equality. Points above the line correspond to subjects for which the temporally weighted method produced a later estimated change-point, while points below the line correspond to subjects for which the temporally weighted method produced an earlier estimated change-point.

To quantify this difference, the subject-level change

$$\Delta\hat{\tau} = \hat{\tau}_{\text{temp}} - \hat{\tau}_{\text{orig}}$$

was computed for each subject. The summary of these differences was:

$$\text{min} = -107, \qquad Q_1 = -55, \qquad \text{median} = 24, \qquad \text{mean} = 12.47, \qquad Q_3 = 71, \qquad \text{max} = 135.$$

A paired t-test was then used to assess whether the temporally weighted method systematically changed the estimated change-point locations. The test produced

$$t = 2.00, \qquad p = 0.047,$$

with a 95% confidence interval for the mean difference given by

$$(0.14,\ 24.79).$$

These results indicate that the temporally weighted method produces significantly larger change-point estimates on average. In other words, incorporating temporal information into the graph construction systematically shifts the detected change-point locations across subjects.

In addition to this average shift, Figure 5 also shows that the temporally weighted estimates are substantially more concentrated than the original estimates. Many subjects have temporally weighted change-points near $\hat{\tau} \approx 116$, suggesting that the proposed modification stabilizes the graph structure and leads to more consistent segmentation across subjects.

Taken together, these findings suggest that temporal weighting does not merely perturb the original graph-based method, but instead changes the effective notion of similarity in a way that produces a different and more stable pattern of detected change-points across subjects.

4.7 Summary

Across all representations, including the scan statistic, PCA trajectory, graph structure, and heatmap, there is consistent evidence of a structural change near $t = 267$. The agreement between statistical, geometric, and visual analyses indicates that the detected change point corresponds to a meaningful shift in the underlying brain activity.

Preliminary analysis showed that different methods identified different candidate change points in the same fMRI run. This suggests that the time series may contain multiple structural changes, and that different methods are sensitive to different aspects of the underlying signal. In particular, graph-based methods detected highly significant candidate change points in the middle and later portions of the run, while other methods produced either multiple candidate changes or different single best split locations.

These early results suggest that comparing methods is important, rather than relying on a single procedure, because each method emphasizes a different definition of what constitutes a change in the data.

5 Discussion

The main lesson from the preliminary analysis is that high-dimensional fMRI data require careful preprocessing before change point methods can be meaningfully compared. The conversion from raw 4D imaging data to a structured time-by-feature matrix was an essential first step. Dimension reduction was also necessary to make multivariate methods practical.

Another important lesson is that disagreement between methods is not necessarily a problem. Instead, it may indicate that the data contain several types of structural changes, and that different procedures are detecting different features of those changes.

References

A Appendix

Additional implementation details, code summaries, and expanded output tables can be added here as the project develops.

A.1 R Code

The data is analyzed using R. All code used in this analysis is included below.

```
library(oro.nifti)
```

```r
library(changepoint)

# 1) Quick check for structure

# original array
nii <- readNIfTI(bold_path, reorient = FALSE)
dim(nii)

# extract the raw array
img <- nii@.Data
dim(img)

# 2) Cleaning

# masking
# begin by keeping any voxel whose value in the first brain volume is not zero
mask <- img[,,,1] != 0
sum(mask) # less voxels

# reshape data
X_vox <- matrix(img[rep(mask, T_len)], nrow = V_len, ncol = T_len)
dim(X_vox) # 300 * V_len

# transpose for statistics
```

```
X <- t(X_vox)
dim(X) # V_len * 300

# check 1st voxel that values are reasonable and changes over time
summary(X[,1])
plot(X[,1], type = "l", main = "First kept voxel over time",
     xlab = "Time point", ylab = "Intensity")

# remove constant/unusable voxel columns
# find sd and mean of each voxel over time
vox_sd <- apply(X, 2, sd)
vox_mean <- colMeans(X)
# keep if finite and positive
keep <- is.finite(vox_sd) & vox_sd > 0 &
  is.finite(vox_mean) & vox_mean > 0

X2 <- X[, keep, drop = FALSE]

dim(X2) # still same but important step

# PCA to get greatest contributions
X2s <- scale(X2) # standardize voxel signals
# build Gram matrix
G <- tcrossprod(X2s)
```

```r
dim(G)

# compute eigen decomposition

eig <- eigen(G, symmetric = TRUE)

# choose number of components

k <- 30

# build the reduced data

vals <- pmax(eig$values[1:k], 0)


Z <- eig$vectors[,1:k] %*% diag(sqrt(vals))


dim(Z) # now it is 300 * k
```

figures/pca_heatmap.png

Figure 4: Heatmap of the first principal components over time. A structural change is visible near $t = 267$.

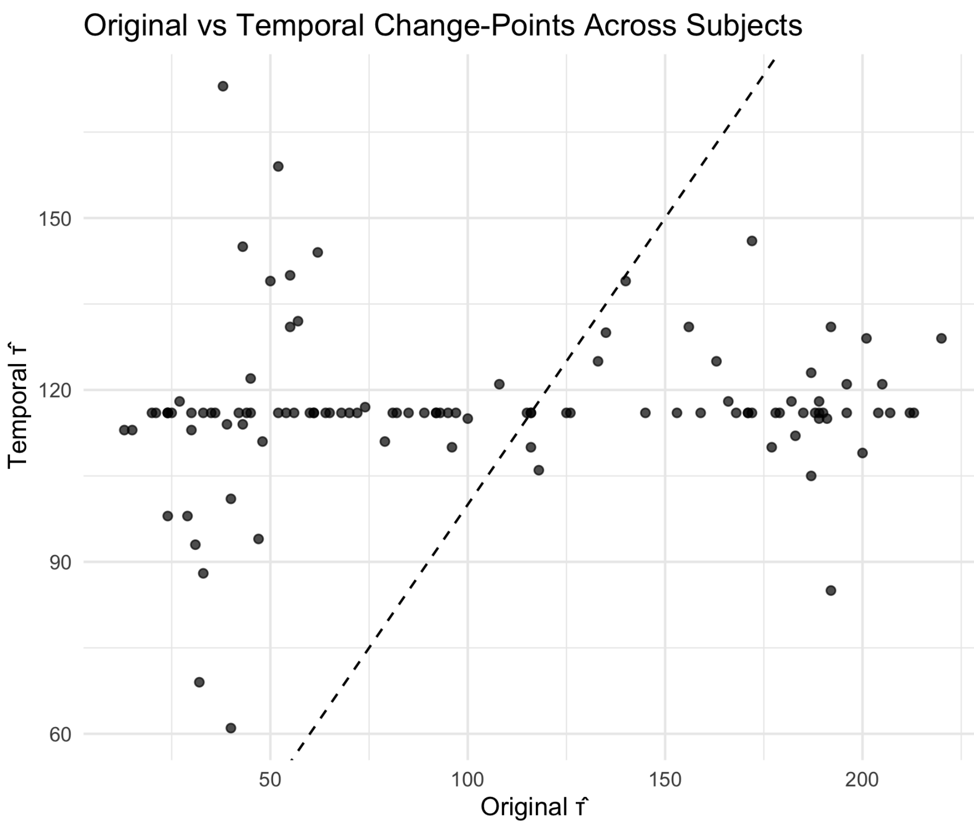

Figure 5: Comparison of subject-level change-point estimates obtained from the original graph-based method and the temporally weighted method with $\lambda = 0.5$. The dashed line represents equality. The concentration of many temporally weighted estimates near a common value indicates increased stability under the proposed method.